

The intrinsic reward is based on the prediction error of states in the environment. This works fairly well on the first pass, but people eventually realized that it runs into another problem, often called the “noisy TV problem” (figure 8.5). It turns out that if you train these agents in an environment that has a constant source of randomness, such as a TV screen playing random noise, the agent will have a constantly high prediction error and will be unable to reduce it. It just stares at the noisy TV indefinitely, since it is highly unpredictable and thus provides a constant source of intrinsic rewards. This is more than just an academic problem, since many real-world environments have similar sources of randomness (e.g., a tree’s leaves rustling in the wind).

Figure 8.5. The noisy TV problem is a theoretical and practical problem where a reinforcement learning agent with a naive sense of curiosity will become entranced by a noisy TV, forever staring at it. This is because it is intrinsically rewarded by unpredictability, and white noise is very unpredictable.



At this point, it seems like prediction error has a lot of potential, but the noisy TV problem is a big flaw. Perhaps we shouldn’t pay attention to the